

Vivek Kumar

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Quantitative Economics & Data Science student with hands-on experience building trading systems, backtesting alpha strategies, and implementing risk management frameworks. Strong foundation in probability, statistical inference, and derivatives theory, with demonstrated ability to identify patterns, develop trading hypotheses, and validate them under realistic market conditions.

EDUCATION

●Birla Institute of Technology, Mesra

Integrated M.Sc in Quantitative Economics and Data Science

2022 – 2027

CGPA: 7.05

RESEARCH & PUBLICATIONS

- **Vivek Kumar**, Dr. Mrinal Jana. “Distribution-free Uncertainty Quantification for Option Pricing Using Residual-space Conformal Prediction.” *CIMA 2026*, **Accepted**. Proceedings to appear in Springer *LNNS* (Paper ID-498, Scopus-indexed).
Developed a novel, distribution-free framework for quantifying prediction uncertainty in derivatives pricing — directly applicable to risk quantification and position sizing where parametric assumptions (e.g., Black-Scholes normality) often fail.

QUANTITATIVE FINANCE & TRADING PROJECTS

●Quantamental Alpha Signal Detection & Paper Trading System

Python, SQL, XGBoost, NetworkX

- Built an end-to-end automated trading pipeline that ingests data from 5+ live sources daily, screens 126+ equities through fundamental filters (D/E ratio, cash flow, promoter pledge), and generates actionable trading signals with a multi-factor conviction score (0–13.5 scale, 8 weighted factors).
- Implemented a 5-indicator technical analysis engine (RSI, MACD, SMA crossovers, OBV, VWAP) with 2×ATR trailing stop-loss for dynamic risk management, and a paper trading simulator with realistic Indian market transaction cost modeling (STT, stamp duty, 0.5% slippage).
- Designed and executed a rigorous 4-test backtesting framework: base-rate connectivity test, post-event excess return analysis (+13.92% alpha at 180 days vs NIFTY benchmark), and walk-forward XGBoost ML optimization (L1/L2 regularized, top feature: pre-event volume Z-score). Achieved **81.8% win rate (9/11)** on a blind 2024–2025 out-of-sample holdout.
- Engineered position sizing using **Fractional Kelly Criterion** (Half-Kelly, 5% max cap) to mathematically optimize capital allocation while controlling for drawdown risk across the paper portfolio.
- Modeled the political-corporate ecosystem as a directed knowledge graph (130+ nodes, 171+ edges) with a fuzzy entity-matching engine (88%+ threshold) to link donor records to BSE scrips; scored connections via a weighted alpha query (Exclusivity 40%, Proximity 30%, Magnitude 30%) with election-cycle multipliers and market-neutral pair-trade identification.

●Regime-Aware Portfolio Optimization via Genetic Algorithms

Python, SciPy, NumPy, Pandas

- Engineered a Regime-Aware Genetic Algorithm from scratch for multi-asset portfolio optimization across a 26-asset global universe spanning equities, fixed income, and commodities.
- Integrated dynamic volatility-based leverage scaling and turnover-aware fitness constraints, achieving a **1.23 Sharpe Ratio** and limiting maximum drawdowns to **-18.5%** — demonstrating disciplined risk/reward optimization.

●Bayesian Statistical Inference & Monte Carlo Simulation

R, Probabilistic Programming

- Applied Metropolis-Hastings MCMC (40,000 iterations) for Bayesian parameter estimation under incomplete data, computing full posterior distributions, credible intervals, and point estimates ($R_0 = 2.72$) — the same class of methods used in stochastic volatility modeling and options pricing.
- Evaluated model convergence using trace diagnostics, autocorrelation analysis, and sensitivity testing — demonstrating rigorous probabilistic reasoning, comfort with stochastic simulation, and disciplined uncertainty quantification.

TECHNICAL SKILLS

Languages: Python, R, SQL

Quantitative & Statistical Methods: Probability & Statistics, Bayesian Inference, Hypothesis Testing, Time Series Forecasting, Regression Analysis, Monte Carlo Simulation, Walk-Forward Backtesting

Trading & Risk: Technical Analysis (RSI, MACD, OBV, ATR), Position Sizing (Kelly Criterion), Event-Driven Strategies, Portfolio Optimization, Sharpe Ratio Analysis, Drawdown Management

Data & Tools: Pandas, NumPy, Scikit-learn, XGBoost, Statsmodels, yfinance, Web Scraping, ETL Pipelines, Data Validation, Git, Docker, Excel

Visualization: Plotly, Matplotlib, Seaborn, Streamlit Dashboards

Relevant Coursework: Econometrics, Multivariate Data Analysis, Time Series Analysis, Probability & Stochastic Processes, Statistical Inference, Financial Economics, Algorithms for Big Data

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

●Events & Public Relations Organizer | *Data Science Summit '23, BITOTSAV '24*

- Spearheaded corporate outreach for DSS 23, successfully pitching sponsors and coordinating cross-functional teams under tight deadlines — demonstrating persuasion, stakeholder management, and ability to perform under pressure.